Juan C. Yamin

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2026–27 Economics Job Market Candidate

Fields: Econometrics, Development
Research interests: Statistical decision theory, experimental design, causal inference

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Dissertation Committee and References

Toru Kitagawa
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Soonwoo Kwon
Assistant Professor of Economics
Brown University
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Jonathan Roth
Professor of Economics
Brown University
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Education

Brown University, Providence, RI
Ph.D. in Economics — Expected May 2027
M.A. in Economics — May 2022

Universidad de los Andes, Bogotá, Colombia
M.Sc. in Economics — March 2019
B.Sc. in Economics — October 2017

Job Market Paper

Poverty Targeting with Imperfect Information

How should antipoverty programs allocate transfers when household income is known only through noisy predictions? I formulate this as a statistical decision problem in which a policymaker chooses nonnegative transfers within a fixed budget to minimize squared deviations of post-transfer income from the poverty line. I show that the standard plug-in rule, which treats predictions as exact, is inadmissible. I then develop a nonparametric empirical Bayes allocation rule that replaces the estimated gaps with posterior mean poverty gaps. The rule's Bayes regret vanishes at least as fast as the mean squared error in estimating posterior mean poverty gaps, despite the budget and nonnegativity constraints. The approach extends, with weaker guarantees, to a policymaker who cares only about how far households remain below the poverty line after transfers and to programs that pay a fixed set of benefit amounts. In simulations using household surveys from nine African countries, the empirical Bayes rule reaches about 1.8 times as many poor people as plug-in OLS with the same budget, and it achieves the same average poverty gap reduction with 6.1% less spending or a 42% smaller training sample.

Other Working Papers

When and How to Pilot: Design Rules for Two-Wave Experiments

Two-Way Effects Models: A Nonparametric Empirical Bayes Approach
with Cole Davis — *Draft available upon request*

Publications

Birds of a Feather Collude Together: Subnational Alignment and Corruption
with Leopoldo Fergusson, Arturo Harker, and Carlos Molina
Forthcoming in the American Political Science Review

Software

cmrdesign (R and Python) Documentation | Code | CRAN | PyPI

Uses pilot data to choose main-wave treatment and control sample shares and report a finite-sample regret certificate. Supports two-arm, multi-arm, and stratified experiments.

Research and Industry Experience

Brown University, Department of Economics June 2023–Jan. 2026

Research Assistant to Soonwoo Kwon, Providence, RI

Conducted simulations and empirical applications on fixed-effect shrinkage, regression discontinuity, and bias-aware inference in regularized regression.

Amazon, Delivery Experience May–Aug. 2025

Economist Intern, Bellevue, WA

Evaluated a nationwide delivery-speed program using difference-in-differences and shift-share IV to estimate direct and geographic spillover effects.

Universidad de los Andes, Department of Economics Aug. 2017–July 2021

Research Assistant to Leopoldo Fergusson and James A. Robinson, Bogotá, Colombia

Built datasets and implemented instrumental-variables, regression-discontinuity, and panel-data methods to study conflict, corruption, and state capacity.

Conference Presentations

Conference on Digital Experimentation (CODE@MIT), Cambridge 2026
When and How to Pilot: Design Rules for Two-Wave Experiments (accepted for presentation)

Midwest Econometrics Group Annual Meeting, Cincinnati 2026
Poverty Targeting with Imperfect Information

World Congress of the Econometric Society, Seoul 2025
Poverty Targeting with Imperfect Information

Advances with Field Experiments, Chicago 2025
When and How to Pilot: Design Rules for Two-Wave Experiments

Awards and Fellowships

Best Student Paper Award, Midwest Econometrics Group 2026
Merit Dissertation Fellowship, Department of Economics, Brown University Fall 2026
International Travel Fund, Graduate School, Brown University 2025
Conference Travel Fund, Graduate School, Brown University 2025

Teaching Experience

Brown University, Teaching Assistant
Applied Econometrics II (Ph.D.), Peter Hull Spring 2026
Applied Econometrics I (Ph.D.), Toru Kitagawa Fall 2024
Using Big Data to Solve Economic and Social Problems (undergraduate), John N. Friedman Fall 2022 and 2023

Universidad de los Andes, Teaching Assistant Aug. 2018–June 2020
Advanced Econometrics (graduate), Raquel Bernal
Political Underpinnings of Prosperity and Poverty (graduate), James A. Robinson
Thinking Problems (undergraduate), Tomás Rodríguez

Academic Service

Organizer, Brown Metrics Coffee 2024–26

Technical Skills and Languages

Tools: R, Python, Stata, MATLAB, SQL, Git, LaTeX
Languages: Spanish (native), English (fluent), French (fluent)